MIS 201 · WEEK 4 · MONDAY 10/12

Ch. 4: AI in Medicine: The Cures Already on the Shelf

Part 1. A predictive AI can only find what its data holds, and the data is thick where money is. Who checked it, and where should the human stand?

Part 2. Every Cure: a nonprofit using AI to find cures that are already sitting in pharmacies.

Don’t forget: the midterm is next Wednesday 10/21

WHERE

In our classroom, on Canvas

WHEN

The whole class period, 2:00 to 3:50

WHAT

Multiple choice, plus two case questions (you can choose two from three)

COVERS

Chapters 1 to 4 and Labs 0 to 4

Bring nothing but yourself: you’ll take it on the classroom machines. Wednesday 10/14 is Lab 4 as usual, and Monday 10/19 is Chapter 5 plus about 30 minutes of review.

You can use the digital textbook if you want. BUT THAT’S IT!

Today

PART 1 · CHAPTER 4 CONCEPTS · ~50 MIN

The data trove and the knowledge graph	~15 min
The shape of the data: when a low score means nobody looked	~10 min
Who checked? External validation	~8 min
Where is the human? In, on, or out of the loop	~12 min
Try it: which concept is doing the work?	~5 min

PART 2 · THE EVERY CURE CASE · ~50 MIN

Watch Penn Medicine's video	~14 min
Background, MATRIX, and Joseph	~6 min
Discuss in groups, then as a class	~30 min
Debrief, wrap-up and attendance	~5 min

BY THE END, YOU CAN

- Explain how ordinary record-keeping becomes a data trove, and who ends up owning the prediction
- Show how a path through a knowledge graph becomes a prediction
- Say why the same low score can mean "no" or "nobody looked"
- Explain why a company's number and an outside number differ
- Say where a human can stand in an AI system, and what each spot costs

PART 1 · CHAPTER 4 · ABOUT 50 MINUTES

AI in Medicine: What can the AI actually see?

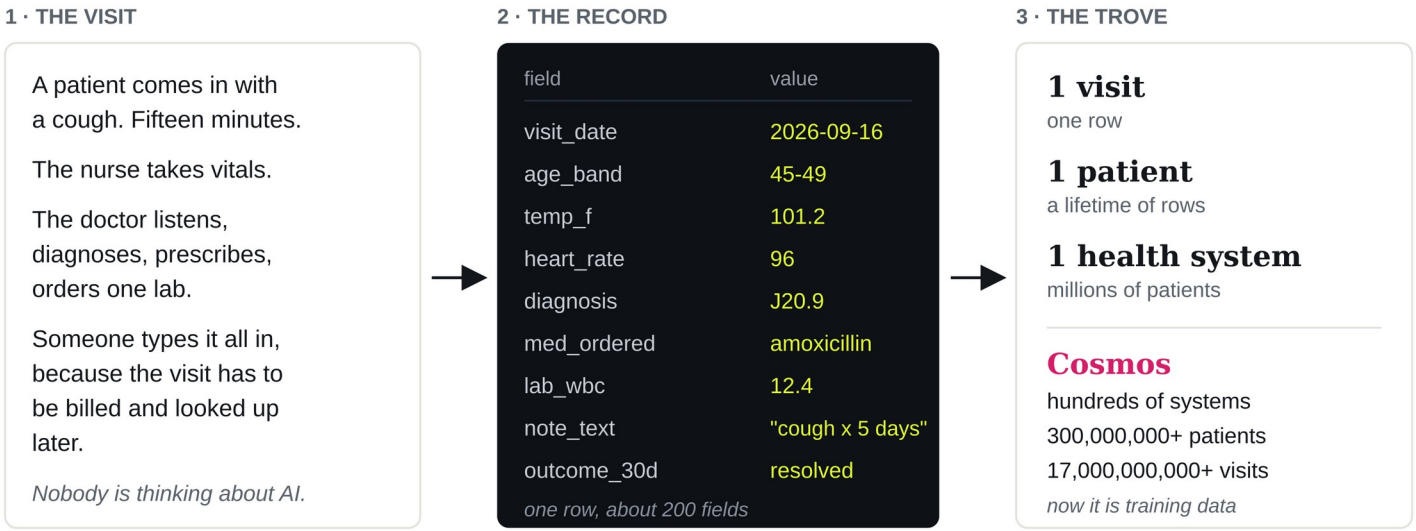
Three questions for the week.

1. What can AI see, what can it never see?
2. Who owns the data it sees, and what does that buy them?
3. Where should the human stand, and who decides?

The point of the chapter: the wins come from the thick parts of the data trove, the failures from the thin parts, and trust comes down to who checked it and where the human stood.

The data trove: one cough, one row

A data trove is a pile of digital records big enough, and clean enough, to train an AI model on. Nobody collected it for AI. A model can only learn what it is fed.



Nobody collected this to build AI. They collected it to bill and to look things up.

The data trove is the asset, the AI model is not

WHAT PEOPLE GET WRONG

They think the AI is the asset.

The data trove is the asset. And nurses, clerks, and patients typed it in.

THE DATA IS THE ASSET

The data's owner owns the prediction.

Bigger trove, better model. Every record someone else adds makes the owner stronger. Economists call that a moat.

ONE PART OF MEDICINE'S DATA TROVE · THE CHART

Enormous, and owned.

About 1 in 8 U.S. hospitals used electronic records in 2009; by 2021, nearly all. Mostly in software from a few private companies.

ANOTHER PART OF MEDICINE'S DATA TROVE · PUBLISHED RESEARCH

Millions of studies, owned by nobody.

Written over decades by people who never met. Today's case, Every Cure, runs on this one.

Questions about any medical AI: ***whose medical records is it learning from****, and* ***who else can get them****? And* ***are any recommendations grounded in evidence****?*

FROM DATA TROVE TO KNOWLEDGE GRAPH

Millions of studies is a pile. Nobody can read a pile.



So Every Cure turns the data into a map they can see.

1. Every drug, protein, cell, and disease becomes a dot.
2. Every published finding becomes a line between two of them.
3. Some parts of the map are crowded. Some are almost empty.
4. Patient records come later, to check the map's best guesses.

First, a map you already use.

What is a knowledge graph, and why use one?

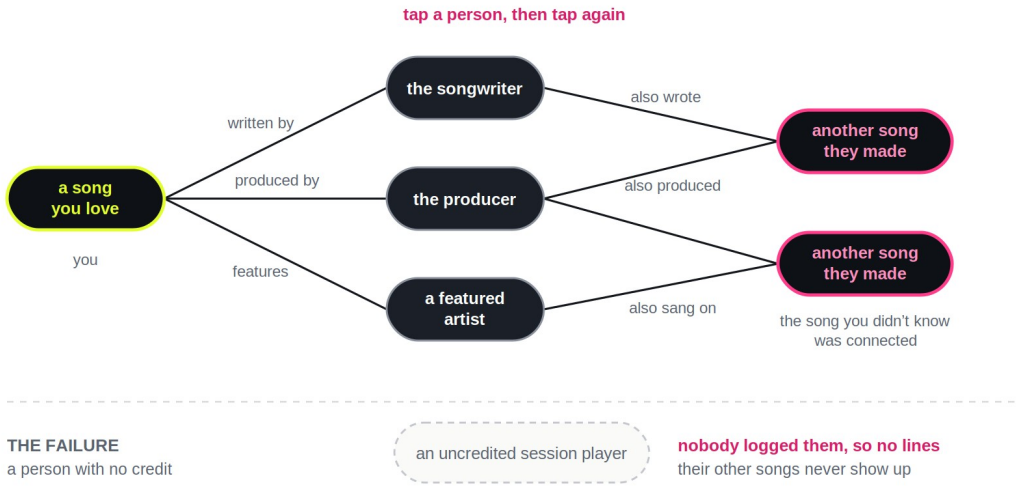
A knowledge graph stores information as dots (nodes) and lines (relationships). Each dot is a thing. Each line is a relationship someone logged: written by, produced by, features.

A path is a route along those lines, like: song → songwriter → another song.

Why a graph? Findings sit in separate places. A graph links them through the data they share.

Where's the AI? The graph is just the data. An AI model learns from connections we already know are good, then scores every other path. The **score is the prediction**, and **the path is its reason**.

Remember: the prediction is built from the data.



You may have seen one: Spotify's SongDNA. Tap a song's writer, then tap again. Every Cure does the same with medicine.

SongDNA: a knowledge graph you can open

Spotify, March 2026. Open a song's Now Playing view, scroll to the SongDNA card, tap in.

THE DOTS

People and songs

Writers, producers, collaborators, the samples a song borrowed, and the covers it inspired.

THE LINES

Wrote, sampled, covered

Tap any creator to see who else they've worked with, then tap one of those. That's hop to hop.

WHO LOGGED IT?

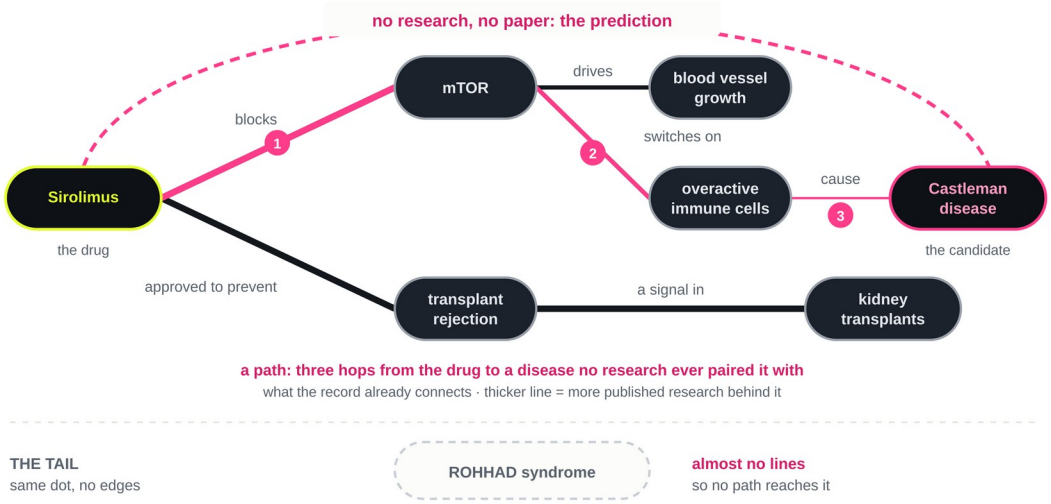
Artists, plus the crowd

Built from what artists' teams send, filled in with community-sourced data. Artist teams can review and fix theirs.

Try it now: pull up a song you love and follow two hops. Did you find a sample, a writer, or a cover you didn't know about? (Premium, in the mobile app.)

A line is only as good as whoever logged it. Wednesday you'll add a false one yourself.

Every Cure does the same thing, with medicine



Thicker line, more research behind it. The pink path gets thinner as it goes.

HOP 1

Sirolimus (a drug) blocks mTOR

Known since the 1990s: why it stops transplant rejection.

HOP 2

mTOR switches on certain immune cells

Documented in immunology research.

HOP 3

Those cells drive one form of Castleman

Drawn by a patient you'll meet in Part 2.

THE ARC

Sirolimus → Castleman disease

No paper ever said this. The path becomes the prediction.

Every solid line is somebody's published research. The AI adds only the dashed arc.

Note: A path (and a prediction) is not a finding. It's a place to look.

WHAT THE AI DOES

Finds paths, scores them, ranks them.

Across millions of papers no person could read. Tens of millions of drug and disease pairs, sorted so the best places to look land at the top.

WHAT THE PEOPLE DO

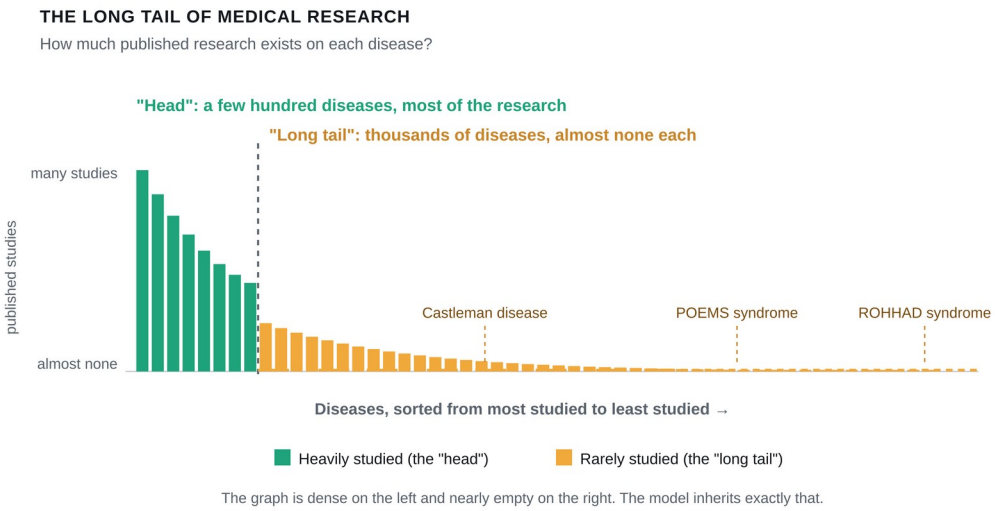
Everything after the score.

Read the score. Pull the papers behind the path. Decide whether the biology holds. Decide whether it's worth a scientist's time, and later, a patient's.

Headline version: "AI discovers a cure." **Accurate:** "AI noticed three published findings line up, and people decided it was worth testing."

The data has a shape: the long tail

The AI can only score paths where someone drew lines. So where are the lines?



GARBAGE IN, GARBAGE OUT

If the records are wrong, the model learns the mistakes as facts. Night shift skips pain scores, so the model learns night patients are rarely in pain.

THE BIGGER PROBLEM: GAPS IN DATA

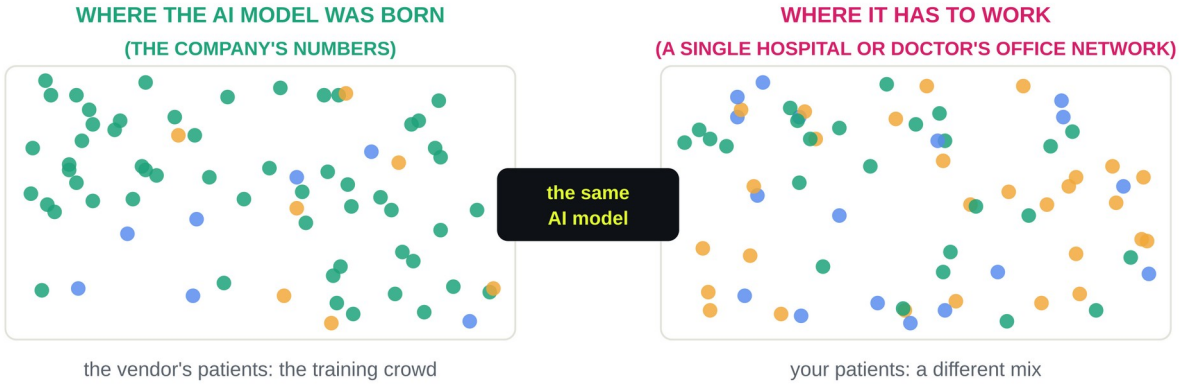
Common diseases in rich countries: lots of data.

Rare diseases nobody funds: almost empty.

The AI model inherits that shape.

Few lines, few paths, weak scores, for every drug. The research gap becomes the prediction gap.

Who ran the models in the first place? Company vs. outside numbers



0.84

accuracy of a heart disease risk score, in the sales deck

Nothing about the model changed. The crowd did.

0.64

the same score, tested on your patients a year later

EXTERNAL VALIDATION

Someone outside the company runs the model on their own data and publishes what happened. In medicine, the difference between a claim and a finding.

DISTRIBUTION SHIFT

A model leaves home: different patients, equipment, habits. Patterns it learned get weaker, so accuracy drops. Nobody has to lie.

Example: Epic's sepsis alert, checked from outside

Sepsis is the body's runaway reaction to an infection, and catching it early saves lives. Epic built an alert for it into its hospital software. In 2021, Michigan Medicine checked it on 38,000 of its own hospital stays.

THE LOUD FAILURE

1 in 5

of all patients set the alert off. Most of them didn't have sepsis.

Alert fatigue.

When most alarms are wrong, nurses learn to click past them, including the right ones.

THE SILENT FAILURE

2 of 3

patients who really had sepsis never set it off.

Nobody finds out.

A patient the alert missed sets off nothing, so no report ever says the alert failed.

Company (proprietary) model: hospitals couldn't look inside.

Where is the human? *In*, *on*, or *out* of the loop

Human in the loop

the AI proposes, a person decides

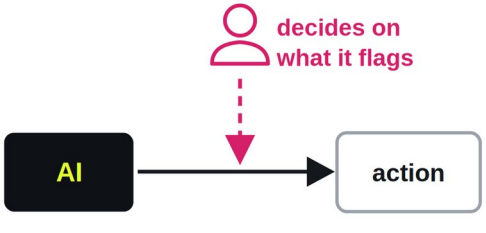


A doctor reads the AI's draft diagnosis and makes the call.

Price: slow, a person on every case

Human on the loop

the AI screens, a person decides what it flags

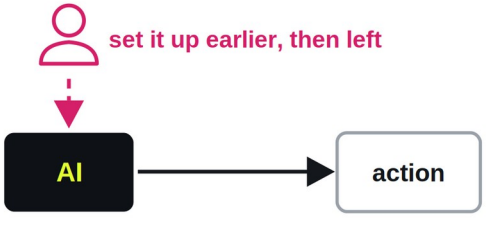


The AI reads every mammogram; a radiologist decides on what it flags.

Price: a person only on the exceptions

Human out of the loop

the AI acts, nobody in the moment



A 2 a.m. symptom chatbot answers with no clinician involved.

Price: nobody there when it's wrong

Each position has a price. ***In the loop*** is slow and expensive, because a person decides every case. ***On the loop***, a person still decides, but only on what the AI flags. ***Out of the loop*** is fast and cheap, and when it goes wrong there is nobody in the room.

WEDNESDAY'S DEBATE

Where is the doctor? *In*, *on*, or *out* in 2026?

A Journal of the American Medical Association viewpoint, August 2026. Four authors argue that for the thinking work of medicine (history, diagnosis, tests, treatment, chronic disease), AI alone will soon beat doctors, and doctors working with AI. Their ask: **regulators should stop requiring a doctor in the loop by default.**

Their strongest evidence (Goh et al., 2024) was a test on paper. 50 doctors read up to six hard written-up patient cases and wrote out their reasoning: likely diagnoses, evidence for and against, next tests. Experts scored each write-up with a points rubric. **Each number is a score, like on an exam: the percent of rubric points a typical write-up earned, not the percent of patients diagnosed correctly.**

74% exam score

Doctors alone

no AI in the room

76% exam score

Doctors using GPT-4

in the loop: AI proposes, doctor decides

~90% exam score

GPT-4 alone

out of the loop

Read the middle number twice. Adding the doctor dragged a 90 down to a 76. Caution: no real patients were involved, and the study's own authors said not to read this as a reason to let AI diagnose alone.

Who's saying that? Two of the four authors on the JAMA article run or fund an AI health company, and disclosed it.

Why could adding a doctor make it worse?

Three habits, and all three are ours, not the machine's.

1

AUTOMATION BIAS

Trusting AI too much.

The system seems reliable, so you stop checking. The radiologist who signs off on "normal" without opening the scan.

2

ALGORITHMIC AVERSION

Trusting AI too little.

Overriding its answer with your own experience when it's more likely right. The JAMA authors' explanation for 76 vs. 90.

Cardiology conference study (Jena et al., *JAMA Internal Medicine*, 2015)

3

ALERT FATIGUE

Tuning it out.

AI fires so often that you learn to click past it. Remember the sepsis alert: a human in the loop becomes a human out of the loop, with extra steps.

On the JAMA view, the doctor isn't the safeguard. The doctor is the noise.

What does other research say? Three hospital trials, three answers

MAMMOGRAMS · SWEDEN

MASAI, 105,000 women

Radiologists moved *on* the loop. The AI sorted scans by risk, and low-risk scans got one radiologist instead of two.

Result: more cancers found, no more false alarms, and much less reading work. **Less human, better result.**

29% more cancers found, no more false alarms, 44% less reading work.

FETAL BRAIN ULTRASOUND · CHINA

1,584 pregnancies, five hospitals

Sonographers stayed in the loop. The AI helped in one of two ways: live, beside the sonographer during the scan, or as a second look afterward.

Result: caught about nine more real problems out of every 100, and live help worked best. **More problems caught, and timing mattered: live help beat a second look.**

PREGNANCY SCANS · ENGLAND

PROMETHEUS trial

Sonographers stayed in the loop, but the AI took over the measuring. As the sonographer scanned, the AI measured the baby automatically.

Result: no confirmed change in accuracy, but scans took about 11 minutes instead of 20. **Same accuracy, ~ half the time, a less exhausted human.**

THE HUMAN GATE The point where a person decides whether an AI's output goes forward. Sometimes the safeguard, sometimes the noise, sometimes there and no longer looking. Which one is a setting you choose and test, task by task.
For which task, tested how, and who answers when it's wrong?

PART 2 · THE CASE · ABOUT 50 MINUTES

Need a break??

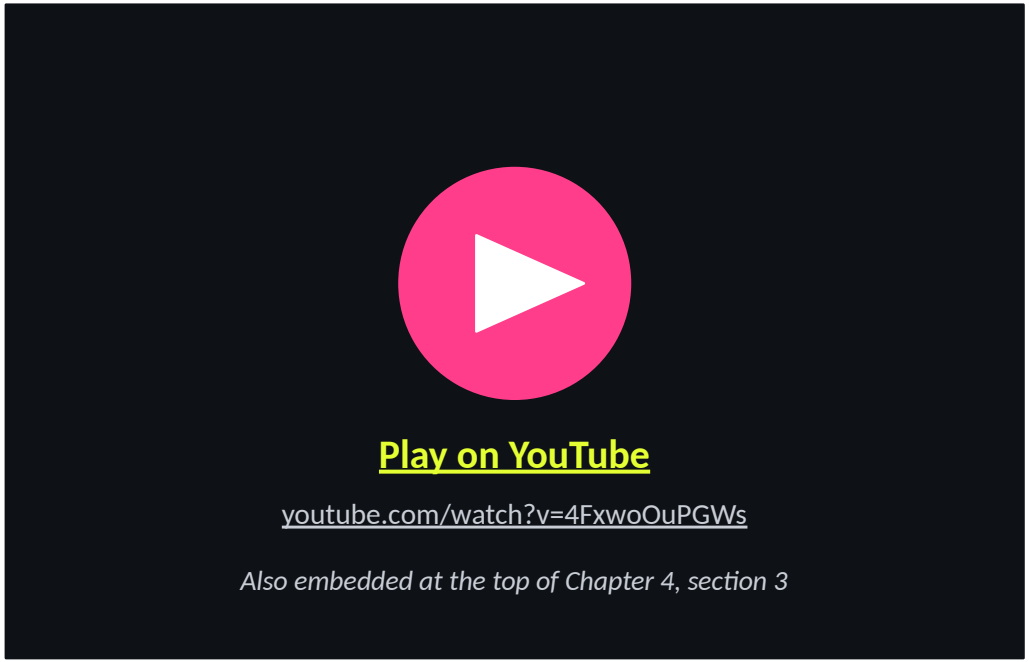
Every Cure: the drug that was already there

A doctor who nearly died of a rare disease found his own cure in a drug that sat in every pharmacy. Then he built a nonprofit to ask how many more are on the shelf.

Video first, then the case taken apart, then you. A story this easy to love is exactly where we run the five questions.

Watch: Penn Medicine on Every Cure

~14 MIN



WATCH FOR THREE THINGS

1. **Fajgenbaum's definition of the system, about eight minutes in.**

   It's the clearest description of a knowledge graph in the chapter.

2. **The moment the AI agrees with him.**

   Who had the idea first?

3. **Who decides to give Joseph chemotherapy.**

   Where is the human in that moment?

Background: the doctor’s own cure

CASE

MED SCHOOL, PENN

His organs begin shutting down. Diagnosis: Castleman disease (his form is called iMCD), a rare disorder where the immune system attacks the body.

FIVE NEAR-DEATHS

The approved treatments fail. It nearly kills him five times.

HIS OWN BLOOD

He studied his own disease and finds an overactive signal.

AN OLD DRUG

Sirolimus, given to transplant patients for decades, switches that signal off. Nobody had tried it for Castleman. He did.

11+ years in remission, on a drug that sat in every pharmacy the whole time he was dying. His next question: how many other cures are already on the shelf? An old generic pill has no patent left, so nobody profits from proving it works for a new disease. Mostly, nobody looks.

Every Cure: MATRIX: the knowledge graph at full size

CASE

8.9 million

dots: drugs, diseases, genes, proteins

33 million

lines, from about 70 public databases

$48.3 million

ARPA-H contract, Feb 2024

up to $76 million

phase two, Feb 2026: 20 lab programs, 10 trials

What it does. Scores every approved drug against every known disease, tens of millions of pairs, and hands the top matches to physicians.

Why AI is needed at all. No team could read millions of papers or check every pair by hand. Before this, repurposing happened by accident, or by one determined doctor.

Public. The scores are online in a heatmap anyone can explore, with a warning: a score marks how plausible a match looks, not proof it works.

Founded 2022. TIME named MATRIX one of the best inventions of 2025.

Joseph: from hospice to remission

CASE

1. Diagnosed with POEMS syndrome, a rare blood disorder. Lost forty pounds in weeks.
2. His oncology team's treatment failed. A stem cell transplant was planned, then called off: too sick.
3. His doctor raised hospice. His partner Tara emailed Fajgenbaum on a Friday afternoon.
4. **POEMS resembles Castleman and multiple myeloma, so myeloma drugs had paths into it. Fajgenbaum had already picked three of those drugs. The ranking agreed.**
5. Responded to treatment within two weeks. Walked again, kidneys recovered, got the transplant, full remission.

THE HARD PART

Chemotherapy is the last thing you give a person who is barely surviving. It could kill them.

The ranking didn't make that call. Doctors who knew him did, and Joseph agreed to it.

A score from an AI model informed it. People decided.

Every Cure: reflect, then discuss

GROUP DISCUSSION

1. **Why does this need AI?** A new drug takes 10 to 15 years and about a billion dollars, and an old generic pill earns nobody a profit. About 4,000 drugs, 18,000 diseases, millions of papers. **Could Every Cure exist without AI? What does the AI do that a team of scientists can't do, and what does Every Cure still need people for?**

2. **Who gets left behind?** Every Cure ranks drug-disease pairs and tests the ones at the top. A rare disease with nine published studies will almost never reach the top, even if a cure is sitting on the shelf. **How does this relate to the data trove and the long tail? Who pays the price for that, and should Every Cure do anything differently for diseases the data barely covers?**

3. **Every Cure looks like beneficial AI.** a nonprofit using AI to find cures nobody else is looking for seems like a good use of AI. **Should we keep, fix, or drop it? What you think? What should we keep about it? What should we fix? What should we drop?**

4. **Your turn: too much data, not enough people.** Pick another place with more records than anyone can read, where AI could do some good: a university's research across departments, a city's 311 complaints, a manufacturer's repair logs, a food bank's donations vs. requests, a hospital's incident reports. **How could AI help with this? What would your data be thin on, and who would get missed?**

1. Groups of 3 or 4, 18 min.
One line each, then settle on an answer to each question.

Nominate a leader.

2. Together, 12 min.
One question at a time. Two or three leaders answer; others add what's different.

Same four questions as the end of Chapter 4, section 3.

Should we keep, fix, or drop Every Cure's AI?

DEBRIEF

A machine read millions of studies no person could, found paths no person had noticed, and a man moved toward hospice is in remission. The data was public, the organization is a nonprofit, and every match goes through people first. If we can't say yes to this, we can't say yes to anything.

KEEP?

The graph, the ranking, and a doctor between every prediction and every patient.

Cost: the record decides who benefits.

FIX?

Publish the misses with the saves. Pay for the trials nobody profits from. Add data on rare diseases.

Cost: money no business model supplies.

DROP?

The hype. "AI found a cure" is a headline. "AI found a candidate, and a trial will tell us" is the truth.

Cost: a less exciting story.

ATTENDANCE BEFORE 4PM PLEASE!

Attendance = your notes from today, uploaded to Canvas right after class.

On Wednesday: Lab 4, Inside the Graph. Build a knowledge graph and train a small language model on it, play the doctor in the loop, then ask Claude what Every Cure should study using its real data. Then the debate: should regulators stop requiring a doctor to sign off on an AI's diagnosis?

On Friday 10/16, 11:59 p.m.: Lab 4 writeup and the Chapter 4 quiz, both on Canvas.

Bring Monday to Wednesday: your side in the debate gets four evidence cards, and two of them are this morning's section 2.5.